

AZURE LOAD BALANCER VS APPLICATION GATEWAY

Design Comparison and Cloud Implementation

Project Documentation

Prepared for Azure Cloud Networking Project

1. Abstract

Cloud computing has become an important technology for developing and deploying modern applications. As the number of users accessing cloud applications increases, applications need to handle large amounts of network traffic without affecting performance, availability, or reliability.

Load balancing is an important technique used in cloud environments to distribute incoming traffic across multiple backend servers or resources. Microsoft Azure provides different services for managing and distributing network traffic, among which Azure Load Balancer and Azure Application Gateway are widely used.

Azure Load Balancer is a Layer 4 load-balancing service that distributes traffic based on information such as IP addresses and ports. It supports TCP and UDP traffic and can distribute traffic across multiple virtual machines and virtual machine scale sets using health probes.

Azure Application Gateway is a Layer 7 web traffic load-balancing service designed for HTTP and HTTPS applications. It can make routing decisions using URL paths and host names. It also supports SSL/TLS termination and can work with Web Application Firewall capabilities.

The project compares both services in terms of networking layer, traffic type, routing, security features, availability, and use cases. The comparison helps in understanding how Azure services can be selected or combined to design scalable, highly available, and secure cloud applications.

2. Introduction

In today's cloud computing environment, applications are expected to handle a large number of users, requests, and data transactions while providing high availability and reliable performance. When many users access an application at the same time, sending all requests to a single server can overload that server and affect application performance.

To solve this problem, cloud platforms provide load-balancing services that distribute incoming network traffic across multiple servers or application instances. Microsoft Azure provides several services for this purpose, among which Azure Load Balancer and Azure Application Gateway are two important solutions.

Azure Load Balancer works primarily at Layer 4 and distributes network traffic using information such as IP addresses and ports. It supports TCP and UDP and is designed for high availability and efficient network-level traffic distribution.

Azure Application Gateway works at Layer 7, the application layer. It understands web requests and can make routing decisions based on URL paths, host names, and other HTTP-related information. It is designed for web applications that need intelligent traffic management.

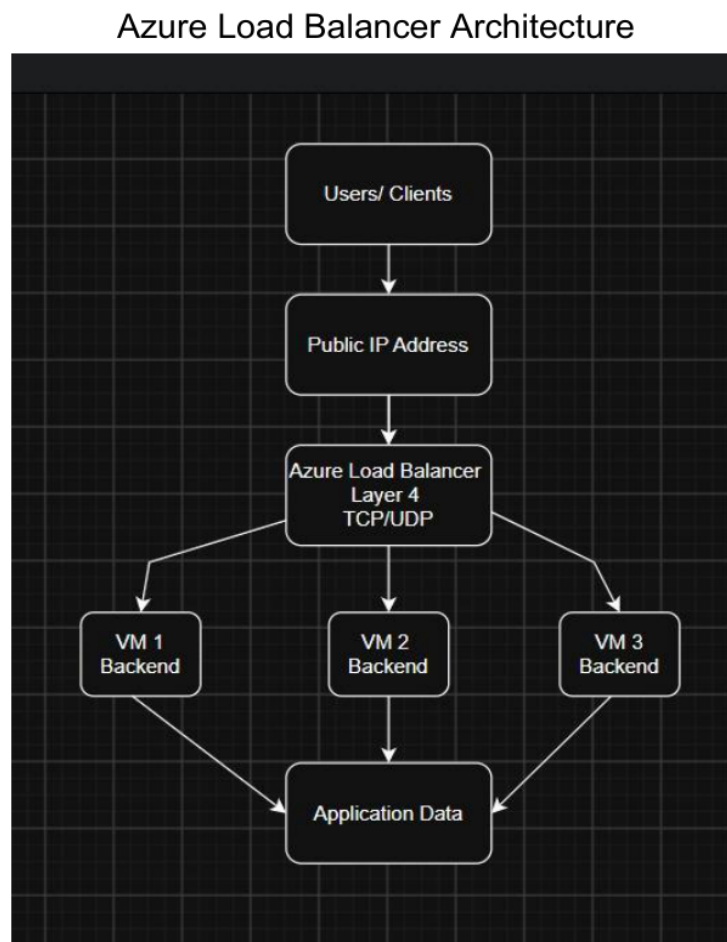
Application Gateway also supports SSL/TLS termination and can integrate with Web Application Firewall capabilities. In modern cloud architectures, both services may be used together, with Application Gateway handling application-level web traffic and Load Balancer handling network-level traffic for backend resources.

The project therefore focuses on understanding the architecture, Azure services, implementation tasks, modules, and differences between these two load-balancing solutions.

3. Architecture

The project contains two architecture designs: one for Azure Load Balancer and another for Azure Application Gateway. The diagrams supplied for the project are included below.

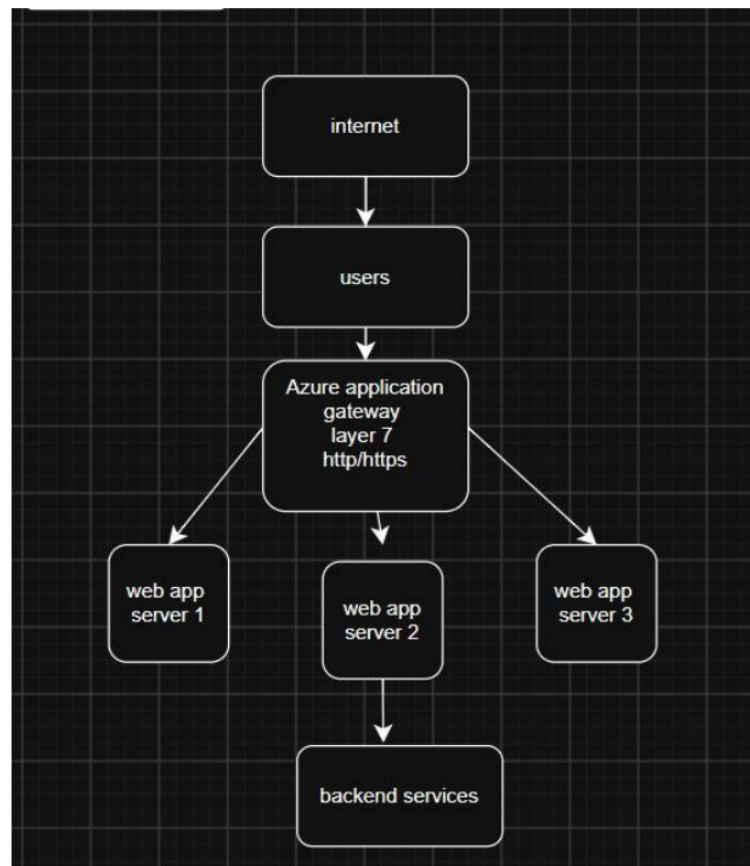
3.1 Azure Load Balancer Architecture



Explanation: Users or clients send traffic to a Public IP Address. The Azure Load Balancer receives the traffic at Layer 4 and distributes TCP/UDP traffic among multiple backend virtual machines. The backend instances provide the application data. Health probes help identify available backend instances so traffic can be directed to healthy resources.

3.2 Azure Application Gateway Architecture

Application gateway



Explanation: Internet users access the application through Azure Application Gateway. The gateway operates at Layer 7 and handles HTTP/HTTPS web traffic. It can inspect web requests and route them to different web application servers. The web application servers then communicate with backend services.

4. Azure Services Required to Implement the Project

Azure Service	Purpose in the Project
Azure Load Balancer	Provides Layer 4 distribution of TCP/UDP network traffic across backend resources.
Azure Application Gateway	Provides Layer 7 HTTP/HTTPS load balancing, application-aware routing, and SSL/TLS termination.
Azure Virtual Network (VNet)	Provides the private network environment in which backend resources and networking components can communicate.
Azure Virtual Machines / VM Scale Sets	Provides multiple backend instances for testing traffic distribution and high availability.
Public IP Address	Provides an internet-facing endpoint for the public load-balancing architecture.
Health Probes	Check the availability of backend instances and help traffic reach healthy resources.
Network Security Group (NSG)	Controls permitted inbound and outbound network traffic for Azure network resources.
Web Application Firewall (WAF)	Can be integrated with Application Gateway to inspect web traffic and provide protection against common web vulnerabilities.
Azure Monitor	Can be used to observe resource health, traffic behavior, and performance during implementation and operation.

5. Design Comparison

Feature	Azure Load Balancer	Azure Application Gateway
Networking Layer	Layer 4	Layer 7
Main Traffic	TCP / UDP	HTTP / HTTPS
Routing Basis	IP address and port	URL path, host name and HTTP information
Application Awareness	Does not inspect HTTP request content	Can inspect web requests
SSL/TLS Termination	Not the main feature	Supported
Web Application Firewall	Not the primary feature	Can integrate with WAF
Typical Backend	VMs / VM Scale Sets and supported resources	Web application servers / backend pools
Main Purpose	Fast network-level traffic distribution	Application-aware web traffic management

6. FIVE TASKS

Task 1 – Azure Network Setup

Create the Resource Group, Virtual Network, subnets, and required network security rules for the project.

Task 2 – Load Balancer Implementation

Create a public Azure Load Balancer, configure a frontend public IP, backend pool, load-balancing rule, and health probe.

Task 3 – Application Gateway Implementation

Create an Application Gateway with frontend access, backend pools, HTTP/HTTPS listener, routing rules, and health checks.

Task 4 – Traffic Routing and Testing

Deploy multiple backend servers and test how requests are distributed through the Load Balancer and Application Gateway.

Task 5 – Comparison and Documentation

Compare Layer 4 and Layer 7 behavior, document observations, capture results, and prepare the final PPT and project report.

7. Project Modules

Module 1 – Network Configuration

Resource group, VNet, subnets, Public IP, and NSG configuration.

Module 2 – Backend Server Module

Creation and configuration of multiple VM/backend instances.

Module 3 – Azure Load Balancer Module

Frontend IP, backend pool, health probe, and Layer 4 load-balancing rule.

Module 4 – Application Gateway Module

Frontend IP, listener, backend pool, routing rules, and Layer 7 web traffic management.

Module 5 – Monitoring and Testing

Traffic tests, backend health verification, availability checks, and comparison of results.

Module 6 – Documentation and Presentation

Architecture explanation, comparison table, project report, Git repository, and PPT.

8. Conclusion

Azure Load Balancer and Azure Application Gateway provide different approaches to traffic distribution. Azure Load Balancer focuses on high-performance Layer 4 network traffic distribution, while Application Gateway provides Layer 7 web traffic management and intelligent routing. Understanding their architecture and capabilities helps in designing scalable and highly available Azure applications. Depending on application requirements, the services can also be used together so that different networking layers are handled by the appropriate Azure component.

9. Final Submission Checklist

- GitHub project link
- Project Abstract
- Project Introduction
- Architecture Diagram with Explanation
- Azure Services Required to Implement the Project
- Five Tasks
- Project Modules
- Project PPT / Presentation
- Final Project Report

Project Title: Azure Load Balancer vs Application Gateway – Design Comparison